How-To Library

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Configuring the Oracle Advanced Security Options in an Oracle Connection in Cloud Data Integration

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Abstract

This article describes how to configure the advanced security options for Oracle Connector in Cloud Data Integration to connect to an Oracle database enabled with advanced security encryption.

Supported Versions

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Overview

You can use Oracle Connector to read data from or write data to an Oracle Database enabled for advanced security.

Oracle Advanced Security provides two important preventive controls to protect sensitive data at the source including transparent data encryption and on-the-fly redaction of display data. The Oracle Advanced Security enabled instance addresses regulatory compliance requirements by protecting sensitive data on the network, on storage media, and within the database from unauthorized disclosure.

To use an Oracle connection to connect to an Oracle database enabled for advanced security, you must configure the Oracle advanced security options in the metadata and runtime properties in the Oracle connection. This article lists the supported Oracle advanced security options that you can configure in an Oracle connection.

Supported Oracle Advanced Security Options

To connect to an Oracle database enabled for advanced security, you can specify the following advanced security options in an Oracle connection:

EncryptionTypes

Specifies one or multiple algorithms to use if Oracle Advanced Security encryption is enabled for the Oracle database.

You can specify the following encryption types for the JDBC and ODBC drivers in an Oracle connection:

- AES256. The driver uses AES with a 256-bit key size.
- AES192. The driver uses AES with a 192-bit key size.
- AES128. The driver uses AES with a 128-bit key size.

For example, you can specify the following encryption type for a JDBC or ODBC driver:

```
EncryptionTypes=AES256
```

EncryptionLevel

Determines whether data is encrypted and decrypted when transmitted over the network between the driver and database server using Oracle Advanced Security encryption.

You can specify the following encryption levels for a JDBC driver in an Oracle connection:

- rejected. Data sent between the driver and the database server is not encrypted or decrypted. The connection fails if the database server specifies `Required`.
- accepted. Data sent between the driver and the database server is encrypted and decrypted if the database server requests or requires it.
- requested. Data sent between the driver and the database server is encrypted and decrypted if the database server permits it.
- required. Data sent between the driver and the database server must be encrypted and decrypted. The connection fails if the database server specifies `Rejected`.

For example, you can specify the following option for the JDBC driver: `EncryptionLevel=accepted`

For an ODBC driver, you must specify values 0, 1, 2, and 3 for the corresponding rejected, accepted, requested, and required encryption level values. For example, you can specify the following option for the ODBC driver: `EncryptionLevel=2`

DataIntegrityLevel

Determines the level of Oracle Advanced Security data integrity used for data sent between the driver and database server. The connection fails if the database server does not have a compatible integrity algorithm.

You can specify the following data integrity levels for a JDBC driver in an Oracle connection:

- rejected. The driver does not enable a data integrity check for data sent between the driver and database server. The connection fails if the database server specifies `required`.
- accepted. The driver enables a data integrity check for data sent between the driver and database server if the database server requests or requires it.
- requested. The driver enables a data integrity check for data sent between the driver and database server if the database server permits it.
- required. The driver performs a data integrity check for data sent between the driver and database server. The database server must have data integrity check enabled. The connection fails if the database server specifies `rejected`.

For example, you can specify the following option for the JDBC driver: `DataIntegrityLevel=requested`

For an ODBC driver, you must specify values 0, 1, 2, and 3 for the corresponding rejected, accepted, requested, and required data integrity level values. For example, you can specify the following option for the ODBC driver: `DataIntegrityLevel=2`

DataIntegrityTypes

Determines the algorithm the driver uses to protect against attacks that intercept and modify data transmitted between the client and server when data integrity is enabled for the Oracle database.

You can specify the following data integrity types for a JDBC or ODBC driver in an Oracle connection:

- MD5. The driver uses the Message Digest 5 (MD5) algorithm.
- SHA1. The driver uses the Secure Hash Algorithm (SHA-1) algorithm.
- SHA1,MD5. If multiple values are specified and Oracle Advanced Security data integrity is enabled using the Data Integrity Level option, the database server determines which algorithm is used based on how it is configured.

For example, you can specify the following option for the JDBC or ODBC driver: `DataIntegrityTypes=MD5`

The following table summarizes the supported Oracle advanced security options for the JDBC and ODBC drivers that you can specify in an Oracle connection:

Property Name	Supported Values for the JDBC Driver	Supported Values for the ODBC Driver
EncryptionTypes	You can specify the following encryption types: - AES256 - AES192 - AES128 Default is the list of all the possible values enabled in the Oracle database.	You can specify the following encryption types: - AES256 - AES192 - AES128 Default is the list of all the possible values enabled in the Oracle database. If you set the EncryptionLevel to 0, the value specified for the EncryptionTypes is ignored.
EncryptionLevel	You can specify the following encryption levels: - rejected - accepted - requested - required Default is accepted.	You can specify the following encryption levels: - 0 - 1 - 2 - 3 Default is 0.
DataIntegrityLevel	You can specify the following data integrity levels: - rejected - accepted - requested - required Default is accepted.	You can specify the following data integrity levels: - 0 - 1 - 2 - 3 Default is 0.
DataIntegrityTypes	You can specify the following data integrity types: - MD5 - SHA1 Default is the list of all possible values enabled in the Oracle database. If you set the DataIntegrityLevel property to rejected, the value specified for the DataIntegrityTypes option is ignored.	You can specify the following data integrity types: - MD5 - SHA1 - SHA1,MD5 Default is SHA1,MD5. If you set the DataIntegrityLevel to 0, the value specified for the DataIntegrityTypes option is ignored.

Configure the Oracle Advanced Security Options in an Oracle Connection

To connect to an Oracle database that is enabled for advanced security, you must configure the Oracle security advanced options for the JDBC and ODBC drivers in the advanced connection properties in the Oracle connection.

When you specify the supported advanced security options in the metadata advanced connection properties, the Secure Agent applies the specified options to the JDBC driver to fetch the metadata. When you specify the supported advanced security options in the runtime advanced connection properties, the Secure Agent applies the specified options to the ODBC driver while running the mapping.

You can specify the advanced security options for the JDBC and ODBC drivers in the following fields in the Oracle connection:

Metadata Advanced Connection Properties

Specify the property for the JDBC driver as a key-value pair. If you want to specify more than one property, separate each property with a semicolon. For example:

```
EncryptionTypes=AES256;EncryptionLevels=accepted;DataIntegrityLevel=accepted;
DataIntegrityTypes=SHA1
```

You can also specify multiple values for a property. In this case, separate multiple values by a comma and enclose all the values within rounded brackets. For example: `EncryptionTypes=(AES128,AES256)`

Runtime Advanced Connection Properties

Specify the property for the ODBC driver as a key-value pair. If you want to specify more than one property, separate each property with a semicolon. For example:

```
EncryptionTypes=AES256;EncryptionLevels=1;DataIntegrityLevel=1;DataIntegrityTypes=SHA1;
DataIntegrityTypes=SHA1
```

You can also specify multiple values for a property. In this case, separate multiple values by a comma and enclose all the values within rounded brackets. For example: `EncryptionTypes=(AES128,AES256)`

Create an Oracle Connection in Cloud Data Integration

Create an Oracle connection and configure the required Oracle properties, including the Oracle advanced security options in Cloud Data Integration.

1. On the **Connections** page in Data Integration, click **New Connection**.

   The **New Connection** page appears.
2. Configure the connection properties.

   The following table describes the connection properties that you must configure:

Property	Description
Connection Name	Name of the Oracle connection.
Description	Description of the connection.
Type	Type of the connection. Select the type of the connection as **Oracle**.

3. Configure the Oracle connection properties:

   The following table describes the Oracle connection properties that you must configure:

Connection property	Description
Runtime Environment	Runtime environment that contains the Secure Agent used to access Oracle.
User Name	User name for the database login. The user name cannot contain a semicolon.
Password	Password for the database login. The password cannot contain a semicolon.

Connection property	Description
Host	Name of the machine that hosts the database server.
Port	Network port number used to connect to the database server. Default is 1521.
Service Name	Service name or System ID (SID) that uniquely identifies the Oracle database. Specify the SID in the following format to connect to Oracle databases: `SID:<ORACLE_SID>`
Schema	Schema used for the Oracle connection.
Code Page	The code page of the database server.
Encryption Method	The method that the Secure Agent uses to encrypt the data exchanged between the Secure Agent and the database server.
Crypto Protocol Version	Cryptographic protocols to use when you enable SSL encryption.
Validate Server Certificate	Validates the certificate that is sent by the database server. If you specify the `HostNameInCertificate` parameter, Secure Agent also validates the host name in the certificate.
Trust Store	The location and name of the trust store file.
Trust Store Password	The password to access the contents of the trust store file.
Host Name in Certificate	Host name of the machine that hosts the secure database. If you specify a host name, the Secure Agent validates the host name included in the connection with the host name in the SSL certificate.
Key Store	The location and the file name of the key store.
Key Store Password	The password for the key store file required for secure communication.
Key Password	The password for the individual keys in the key store file required for secure communication.
Connection Retry Period	Number of seconds the Secure Agent attempts to reconnect to the Oracle database if the connection fails. If the Secure Agent cannot connect to the database in the retry period, the operation fails. Used for all operations. Default is 0.
Metadata Advanced Connection Properties	The optional properties for the JDBC driver to fetch the metadata. If you specify more than one property, separate each key-value pair with a semicolon. To connect to an Oracle database enabled for advanced security, specify the advanced security encryption options.
Runtime Advanced Connection Properties	The optional properties for the ODBC driver for running mappings. If you specify more than one property, separate each key-value pair with a semicolon. To connect to an Oracle database enabled for advanced security, specify the advanced security encryption options.

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